

Monetary Economics and Policy

Pierpaolo Benigno

Chapter 6: An AS-AD Graphical
Analysis



Prepared by Marcos Consuegra

What We Will We Learn

- ▶ We derive simplified expressions for the **Aggregate Demand** and **Aggregate Supply** equations of the New-Keynesian Model, highlighting their slopes, intuition, and policy shifters.
- ▶ We introduce the **frictionless (Wicksellian) real interest rate** r_t^e and use it to interpret monetary policy stabilization in terms of price stability and output-gap elimination.
- ▶ We analyze how different shocks—**productivity, markup, fiscal, preference, and expectation shocks**—shift AD and AS differently, producing or eliminating policy trade-offs.
- ▶ We identify when monetary policy can jointly stabilize inflation and the output gap, and when it faces an unavoidable **trade-off**.

Outline

- 1 Introduction
- 2 Outline of the Results
- 3 Productivity Shocks
- 4 Markup Shock
- 5 Fiscal (Public Spending) Shock
- 6 Preference Shock
- 7 Disanchoring of Long-Run Price Expectations
- 8 Key Takeaways

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The Two Core Equations of the New Keynesian Model

1. New Keynesian Aggregate Supply (NK Phillips Curve)

$$\pi_t - \pi = \kappa (\hat{Y}_t - \hat{Y}_t^n) + \beta E_t(\pi_{t+1} - \pi),$$

where:

$$\kappa = \frac{1 - \alpha}{\alpha} \frac{(1 - \alpha\beta)(\sigma^{-1} + \eta)}{1 + \theta\eta}.$$

- ▶ Inflation increases with the **output gap**.
- ▶ Future output gaps and expected inflation affect current inflation.
- ▶ Sticky prices ($\alpha > 0$) → **AS is upward sloping**.

The Two Core Equations of the New Keynesian Model

2. New Keynesian Aggregate Demand (IS Equation)

$$\hat{Y}_t = \hat{G}_t + E_t(\hat{Y}_{t+1} - \hat{G}_{t+1}) - \sigma [\hat{l}_t - E_t(\pi_{t+1} - \pi) + E_t(\hat{\xi}_{t+1} - \hat{\xi}_t)] .$$

- ▶ Output depends negatively on the **real interest rate**.
- ▶ Expectations about future rates affect output *today*.
- ▶ Fiscal demand and preference shocks shift *AD* directly.

AS-AD Framework: Simplifying Assumptions

Assumptions:

- ▶ **Two-period** perfect-foresight economy: short run t , long run $t + 1$.
- ▶ In the long run, prices are fully flexible and output equals the natural level:

$$\hat{Y}_{t+1} = \hat{Y}_{t+1}^n.$$

- ▶ Policy anchors the long-run price level:

$$p_{t+1} = p^*.$$

Normalizations:

- ▶ Inflation target set to zero: $\pi = 0$.
- ▶ Initial price level fixed at the long-run anchor:

$$p_{t-1} = p^*.$$

From NK IS Equation to the AD Representation

Starting point: NK Aggregate Demand (IS Equation)

$$\hat{Y}_t = \hat{G}_t + E_t(\hat{Y}_{t+1} - \hat{G}_{t+1}) - \sigma [\hat{i}_t - E_t(\pi_{t+1} - \pi) + E_t(\hat{\xi}_{t+1} - \hat{\xi}_t)] .$$

Step 1: Perfect-foresight assumption

$$E_t(\hat{Y}_{t+1}) = \hat{Y}_{t+1}, \quad E_t(\hat{G}_{t+1}) = \hat{G}_{t+1}, \quad E_t(\hat{\xi}_{t+1}) = \hat{\xi}_{t+1} .$$

Step 2: Long-run equilibrium conditions

$$\underbrace{\hat{Y}_{t+1}}_{\text{future output}} = \underbrace{\hat{Y}_{t+1}^n}_{\text{natural level}}, \quad \underbrace{\pi_{t+1}}_{\text{future inflation}} = p_{t+1} - p_t = \underbrace{p^* - p_t}_{\text{price-level anchor}} .$$

Step 3: Substitute into NK IS and simplify

$$\hat{Y}_t = \hat{G}_t + (\hat{Y}_{t+1}^n - \hat{G}_{t+1}) - \sigma [\hat{i}_t + (p_t - p^*) + (\hat{\xi}_{t+1} - \hat{\xi}_t)] .$$

AD Equation: Short-Run Relationship

AD Equation under Perfect Foresight:

$$\hat{Y}_t = \hat{G}_t + \hat{Y}_{t+1}^n - \hat{G}_{t+1} - \sigma \left(\hat{i}_t + p_t - p^* + \hat{\xi}_{t+1} - \hat{\xi}_t \right) \quad (6.1)$$

Properties:

- ▶ Negative short-run relationship between output $\hat{Y}_t$ and price level p_t $\left(\frac{\partial \hat{Y}_t}{\partial p_t} = -\sigma \right)$.
- ▶ Higher p_t raises the real interest rate $\Rightarrow$ postponing consumption $\Rightarrow \hat{Y}_t \downarrow$.
- ▶ Figure 1 illustrates the downward-sloping schedule.

AD Shifters:

- ▶ Exogenous shocks: $\hat{G}_t, \hat{G}_{t+1}, \hat{\xi}_t, \hat{\xi}_{t+1}$.
- ▶ Long-run changes in natural output: $\hat{Y}_{t+1}^n$.
- ▶ Monetary policy shifts via $\hat{i}_t$ and long-run price target p^* .

From NKPC to the Short-Run Aggregate Supply Curve

Starting point: New Keynesian Phillips Curve

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t(\pi_{t+1} - \pi).$$

Step 1: Perfect-foresight, two-period setting

$$E_t(\pi_{t+1}) = \pi_{t+1}, \quad \pi_t = p_t - p_{t-1}, \quad \pi_{t+1} = p_{t+1} - p_t.$$

Step 2: Long-run anchors and inflation target

$$p_{t+1} = p^*, \quad p_{t-1} = p^*, \quad \pi = 0.$$

Substitute into NKPC:

$$p_t - p^* = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta(p^* - p_t).$$

Short-run AS relationship:

$$p_t - p^* = \frac{\kappa}{1 + \beta} (\hat{Y}_t - \hat{Y}_t^n).$$

AS Equation: Short-Run Relationship

AS Equation under Perfect Foresight:

$$p_t - p^* = \frac{\kappa}{1 + \beta} (\hat{Y}_t - \hat{Y}_t^n) \quad (6.2)$$

Properties:

- ▶ Positive short-run relationship between price level p_t and output $\hat{Y}_t$ $\left(\frac{\partial p_t}{\partial \hat{Y}_t} = \frac{\kappa}{1+\beta} \right)$.
- ▶ Higher output increases marginal costs, driving prices upward.
- ▶ Figure 1 depicts the upward-sloping curve. Note that the AS passes through $\hat{Y}_t = \hat{Y}_t^n$ and $p_t = p^*$.

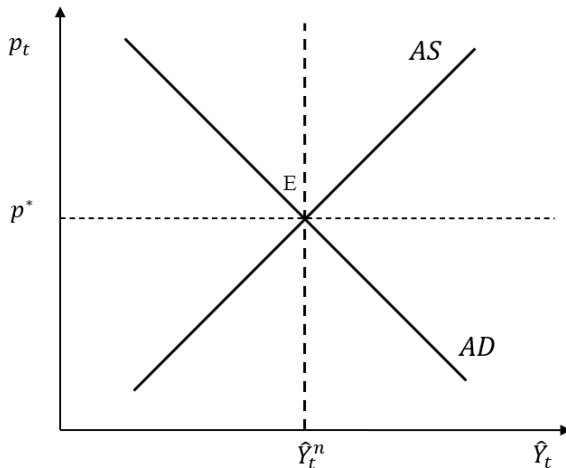
AS Shifters:

- ▶ Only short-run shocks affecting the natural level $\hat{Y}_t^n$ shift the schedule (e.g., via productivity, government spending and markup shocks).
- ▶ Long-run monetary policy modifies p^* but leaves p_{t-1} unaffected.

Baseline Equilibrium

Short-Run Equilibrium E :

- ▶ Intersection of AD (6.1) and AS (6.2).
- ▶ AS equation passes through $\hat{Y}_t = \hat{Y}_t^n$ and $p_t = p^*$.



Natural and Efficient Output: Short and Long Run

Short run (t):

$$\hat{Y}_t^e = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_t + \frac{\sigma^{-1}}{\sigma^{-1} + \eta} \hat{G}_t, \quad (6.3)$$

$$\hat{Y}_t^n = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_t + \frac{\sigma^{-1}}{\sigma^{-1} + \eta} \hat{G}_t - \frac{1}{\sigma^{-1} + \eta} \hat{\mu}_t, \quad (6.4)$$

$$(6.5)$$

Long run ($t + 1$):

$$\hat{Y}_{t+1}^e = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_{t+1} + \frac{\sigma^{-1}}{\sigma^{-1} + \eta} \hat{G}_{t+1}, \quad (6.6)$$

$$\hat{Y}_{t+1}^n = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_{t+1} + \frac{\sigma^{-1}}{\sigma^{-1} + \eta} \hat{G}_{t+1} - \frac{1}{\sigma^{-1} + \eta} \hat{\mu}_{t+1}. \quad (6.7)$$

- ▶ $\eta > 0$: inverse Frisch elasticity; $\sigma > 0$: IES.
- ▶ Assumption: $\hat{Y}_{t+1}^n = \hat{Y}_{t+1}^e$ (by setting $\hat{\mu}_{t+1} = 0$ inefficient shocks are transitory).

Frictionless Real Rate of Interest

Concept

- ▶ The *frictionless real rate* r_t^e is the real interest rate consistent with **price stability** ($p_t = p^*$) and with achieving the **efficient level of output** in the short run.
- ▶ It is driven by efficient (non-distortionary) shocks: productivity, public spending, and preference shocks.

Frictionless real rate of interest

$$r_t^e = \frac{1}{\sigma} \left(\hat{Y}_{t+1}^e - \hat{Y}_t^e - (\hat{G}_{t+1} - \hat{G}_t) \right) + (\hat{\xi}_t - \hat{\xi}_{t+1}).$$

AD equation in terms of r_t^e

$$\hat{Y}_t - \hat{Y}_t^e = -\sigma \left(\hat{i}_t - (p^* - p_t) - r_t^e \right).$$

- ▶ If the real rate satisfies $\hat{i}_t - (p^* - p_t) = r_t^e$, the economy attains efficient output and stable prices.
- ▶ This is the essence of the **Wicksellian policy principle**.

Frictionless Real Rate of Interest

Equivalent expression in primitive shocks

$$r_t^e = \frac{1 + \eta}{1 + \sigma\eta} (\hat{A}_{t+1} - \hat{A}_t) - \frac{\eta}{1 + \sigma\eta} (\hat{G}_{t+1} - \hat{G}_t) - (\hat{\xi}_{t+1} - \hat{\xi}_t). \quad (6.8)$$

Implications

- ▶ **Productivity improvements** $\Rightarrow r_t^e$ rises. Efficient output grows faster $\rightarrow$ desired consumption smoothing shifts demand forward.
- ▶ **Government spending increases** (future vs. today) affect r_t^e through wealth and labor-supply channels.
- ▶ **Preference shocks** shift r_t^e through intertemporal demand for consumption.

Policy Insight

- ▶ r_t^e is the real rate that the central bank would implement to keep inflation at target *and* maintain efficient output.
- ▶ Related modern concept: **R-star** commonly used by central banks.

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Outline of the Results

- ▶ Under **efficient shocks** (productivity, public spending), **no policy trade-off** arises: stabilizing prices and the efficient output gap is jointly attainable.
- ▶ The **required interest rate adjustment** depends on the shock type:
 - ▶ temporary vs. permanent shocks,
 - ▶ news about future shocks.
- ▶ Fluctuations in the **frictionless interest rate** r_t^e serve as an anchor for policy guidance.
- ▶ **Inefficient shocks** (e.g. markup shocks) generate a policy **trade-off**, and r_t^e can no longer serve as the sole reference for the policy rate.

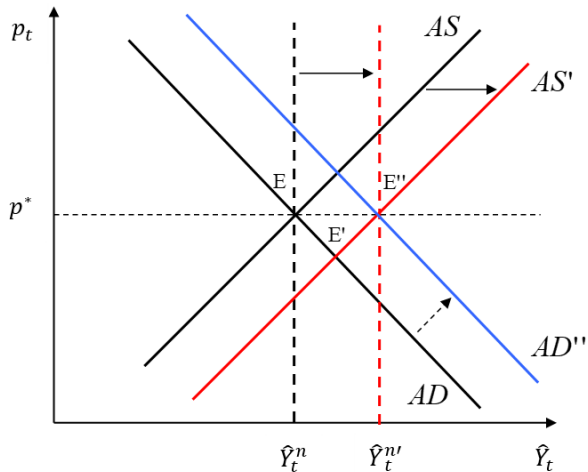
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Temporary Productivity Shock

Scenario: A temporary productivity increase at t : ($\hat{A}_t \uparrow$).

Implications:

- ▶ $\hat{Y}_t^n \uparrow$, see (6.4).
- ▶ **AS shifts downward to AS' :** Higher $\hat{A}_t$ reduces real marginal costs; adjusting firms lower prices.
- ▶ Equilibrium E' is reached along the **AD**: A decrease in short-run prices lowers real interest rate, and raises consumption.



Monetary Policy Response

Without intervention, **equilibrium** (E') is reached.

- ▶ Price level is below initial equilibrium, output is higher but remains below the efficient and natural levels; output gap is negative.
- ▶ Note that efficient and natural output increase in the same proportion.

Achieving **equilibrium** (E'') with price stability and closing the output gap:

- ▶ A lower nominal rate $\hat{i}_t$ shifts AD upward to AD'' .
- ▶ The new intersection E'' features both $p_t = p^*$ and $\hat{Y}_t = \hat{Y}_t^n = \hat{Y}_t^e$.

Wicksell interpretation:

- ▶ The productivity shock decreases the frictionless real rate r_t^e , see (6.8).
- ▶ The policy rate must fall to track r_t^e to achieve price stability and eliminate the negative output gap.

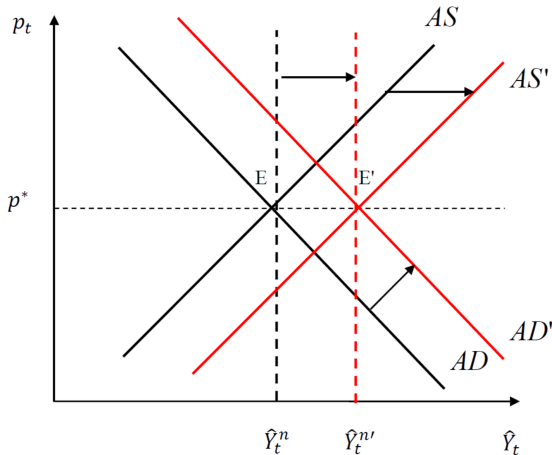
Permanent Productivity Shock

Scenario:

- ▶ Permanent increase in productivity: $\hat{A}_t \uparrow$ and $\hat{A}_{t+1} \uparrow \Rightarrow \hat{Y}_t^n$ and $\hat{Y}_{t+1}^n \uparrow$.

Implications:

- ▶ **AS shifts downwards to AS':**
Lower marginal costs (same as temporary shock).
- ▶ **AD shifts upwards to AD':**
 $\hat{Y}_{t+1}^n$ rises. Households anticipate higher future income and smooth consumption by increasing demand today.



Policy Response

Equilibrium E' :

- ▶ The AD curve shifts exactly to intersect the new AS curve at E' .
- ▶ Result: Output increases to the new natural level ($\hat{Y}_t = \hat{Y}_t^n$), coinciding with the efficient level.
- ▶ Prices remain stable ($p_t = p^*$) without intervention.

Why no policy intervention?

- ▶ **Real Rate Stability:** A permanent gain in productivity does *not* change the frictionless real interest rate (r_t^e), as shown in (6.8).
- ▶ Since $\hat{Y}_t^n$ and $\hat{Y}_{t+1}^n$ increase in the same proportion, the "growth rate" of natural output is unchanged.

Conclusion:

- ▶ The economy achieves price stability and closes the output gap automatically.

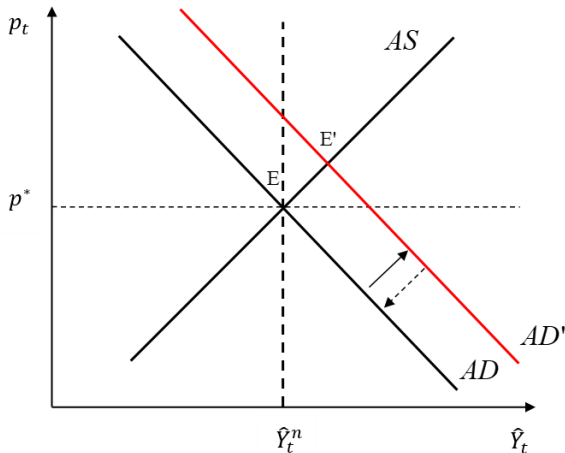
Optimism or "News" on Future Productivity

Scenario:

- ▶ Expected increase in future productivity: $\hat{A}_{t+1} \uparrow$ (but $\hat{A}_t$ remains constant).

Implications:

- ▶ **AS unchanged:** Current productivity is unchanged, so current marginal costs are unaffected.
- ▶ **AD shifts upwards to AD':** Households anticipate higher future income ($\hat{Y}_{t+1}^n \uparrow$). To smooth consumption, they increase demand today.



Policy Response

The New Equilibrium (E'):

- ▶ Intersection of the original AS and the new AD' .
- ▶ Output expands beyond the natural level ($\hat{Y}_t > \hat{Y}_t^n$), and the efficient level.
- ▶ **Inflationary:** High demand drives up real wages and marginal costs, forcing firms to raise prices.

Restoring price stability and closing output gap in **equilibrium** (E''):

- ▶ The central bank must **raise the nominal interest rate**.
- ▶ This restricts demand, shifting the AD curve back down to the initial point E .

Wicksellian Interpretation:

- ▶ Because the frictionless real rate (r_t^e) has increased, as in (6.8), the policy rate should follow.

Productivity Shocks — Key Takeaways

- ▶ Monetary policy can always stabilize **both inflation and the output gap** under productivity shocks.
- ▶ The **direction** of the interest rate movement depends on the nature of the shock:
 - ▶ **Temporary positive shock**: policy should be expansionary (rate cut).
 - ▶ **Permanent productivity improvement**: policy unchanged.
 - ▶ **Expected future productivity growth**: policy should tighten (rate hike).
- ▶ The **Wicksellian natural rate** is the benchmark: the policy rate must adjust in the same direction as changes in r_t^e .

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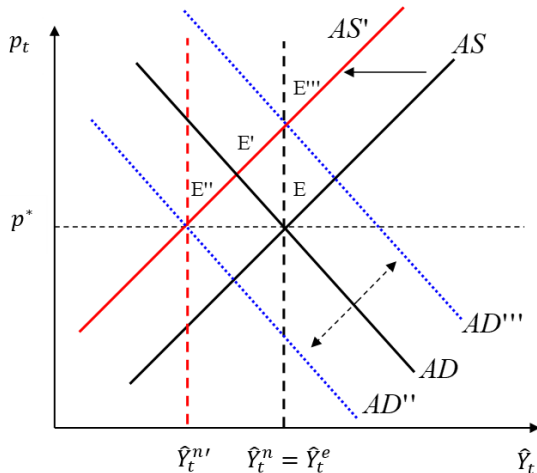
Markup Shock

Scenario

- ▶ Temporary increase in the markup: $\hat{\mu}_t \uparrow$.

Implications:

- ▶ $\hat{Y}_t^e$ fixed; $\hat{Y}_t^n \downarrow$, see (6.3) & (6.4).
- ▶ **AS shifts upwards to AS'** : firms raise prices driven by the cost-push shock.
- ▶ **AD unchanged**, adjustment along the schedule due to higher real rate reducing consumption.
- ▶ **Stagflation**: In equilibrium E' stagnation and inflation. Output falls and prices increase.



Stagflation: Policy Trade-Off

A markup shock creates a direct policy conflict: stabilizing prices and maintaining efficient output cannot be achieved simultaneously.

Option A: Stabilize Prices (E'')

- ▶ Raise nominal interest rate ($i_t \uparrow$).
- ▶ Aggregate demand shifts down ($AD \rightarrow AD''$).
- ▶ Inflation returns to target, but output contracts further.

Option B: Reach Efficient Output (E''')

- ▶ Lower nominal interest rate ($i_t \downarrow$).
- ▶ Aggregate demand shifts up ($AD \rightarrow AD'''$).
- ▶ Output returns to $\hat{Y}_t^e$, but inflation rises.

Wicksellian implication:

- ▶ Frictionless real rate r_t^e is unchanged, so it **no longer guides optimal policy**.

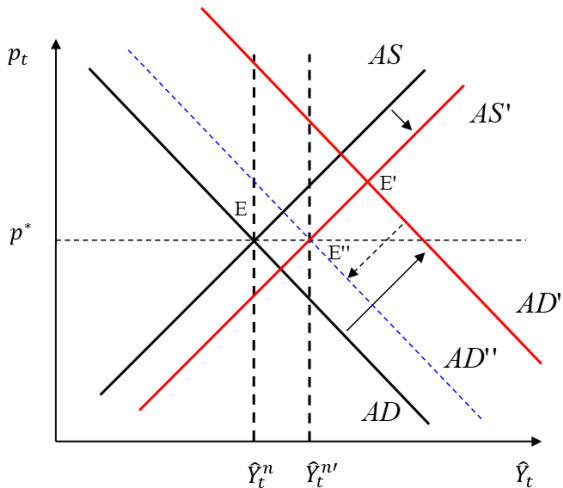
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Temporary Public Spending Shock

Scenario: A temporary increase in public spending at t : ($\hat{G}_t \uparrow$).

Implications:

- ▶ $\hat{Y}_t^e \uparrow$; $\hat{Y}_t^n \uparrow$, see (6.3) & (6.4).
- ▶ **AD shifts upward to AD' :** Higher $\hat{G}_t$ directly expands demand for given consumption.
- ▶ **AS shifts downward to AS' :** Higher natural output $\hat{Y}_t^n$ shifts AS downward.



Policy Response

Equilibrium (E'):

- ▶ AD shift dominates the AS shift: Prices rise and output expands.
- ▶ Equations (6.1), (6.2) and (6.4) imply a **positive output gap** unless $\eta = 0$.

Restoring the efficient equilibrium (E''):

- ▶ A higher nominal rate $\hat{i}_t$ shifts AD' downward to AD'' .
- ▶ The new intersection E'' features both $p_t = p^*$ and $\hat{Y}_t = \hat{Y}_t^n = \hat{Y}_t^e$.

Wicksell interpretation:

- ▶ Because the frictionless real rate (r_t^e) has increased, the policy rate should follow.
- ▶ **Note:** If $\eta = 0$, r_t^e remains constant; AD and AS adjust proportionally, automatically closing the output gap and keeping prices stable without policy intervention.

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Preference Shock

Scenario: Households become more patient: $\hat{\xi}_t \downarrow$ or $\hat{\xi}_{t+1} \uparrow$.

Implications:

- ▶ Households more patient today $\rightarrow$ want to save more.
- ▶ Frictionless equilibrium: $r_t^e \downarrow$, see (6.8) to clear excess saving since efficient output unchanged ($\hat{Y}_t^e$).
- ▶ With sticky prices: AD shifts downward $\rightarrow \hat{Y}_t \downarrow, p_t \downarrow$ (along the unchanged AS).
- ▶ Output falls below efficient level.
- ▶ Prices decline along AS curve.

Preference Shock — Policy Response

Monetary Policy Response:

- ▶ Cut nominal interest rate $\hat{i}_t$ proportionally to the fall in r_t^e .
- ▶ AD shifts back upward $\rightarrow$ restores $p_t = p^*$ and closes output gap.

Key Insights:

- ▶ Preference shocks affect only AD , similar to expected productivity news.
- ▶ Movements in the frictionless rate r_t^e guides policy adjustment.

Liquidity trap preview:

- ▶ Large fall in $\hat{\xi}_t$ can push $r_t^e < 0$; if zero lower bound binds, stabilizing prices and output simultaneously may be infeasible (see Chapter 9).

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Disanchoring Expectations

Assumption: We relax the assumption that long-run price expectations are rigidly anchored at p^* . We distinguish between:

- ▶ $E_t^d p_{t+1}$: Consumers' expectations (affects Demand).
- ▶ $E_t^s p_{t+1}$: Firms' expectations (affects Supply).

Modified AD Equation (Consumer Expectations):

$$\hat{Y}_t - \hat{Y}_t^e = -\sigma \left(\hat{i}_t - (E_t^d p_{t+1} - p_t) - r_t^e \right)$$

Modified AS Equation (Firm Expectations):

$$p_t = \frac{1}{1 + \beta} p^* + \frac{\kappa}{1 + \beta} (\hat{Y}_t - \hat{Y}_t^n) + \frac{\beta}{1 + \beta} E_t^s p_{t+1}$$

Disanchoring: Consumer Inflation Expectations

Scenario:

- ▶ Consumers adjust long-run price expectations downward ($E_t^d p_{t+1} \downarrow$).

Implications:

- ▶ Lower expected inflation leads to an increase in the real interest rate (for a given nominal i_t).
- ▶ Households increase savings and postpone consumption.
- ▶ The AD curve shifts **downward** from the initial equilibrium.
- ▶ Output and prices both fall ($\hat{Y}_t \downarrow, p_t \downarrow$) along the unchanged AS leading to a **potential deflationary spiral**.

Policy:

- ▶ The central bank should consider lowering the nominal rate to stimulate demand and shift AD back.

Disanchoring: Firm Inflation Expectations

Scenario:

- ▶ Firms anticipate a future increase in the price level ($E_t^s p_{t+1} \uparrow$).

Implications:

- ▶ Strategic pricing: firms incorporate higher future prices into current decisions.
- ▶ The AS curve shifts **upward** from the initial equilibrium.
- ▶ Current prices rise while output contracts ($\hat{Y}_t \downarrow, p_t \uparrow$), leading to a **potential inflationary spiral**.

Policy:

- ▶ **Stagflation:** The economy faces a trade-off (similar to a Markup Shock).

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Main Takeaways

- ▶ With **efficient shocks** (productivity, public spending, preference): aligning the policy-controlled real rate

$$\hat{i}_t - (p^* - p_t),$$

with the efficient real rate r_t^e delivers price stability *and* closes the efficient output gap.

- ▶ With **inefficient shocks** (e.g. markup shocks): the efficient real rate r_t^e is no longer an appropriate guide. Policymakers face a *trade-off* between stabilizing inflation and closing the output gap.
- ▶ **Expectations matter**: a disanchoring of long-run price expectations $E_t p_{t+1}$ can be destabilizing. This reinforces the need for monetary policy to firmly anchor expectations.

9 Appendix

Shock-by-Shock Logic

Shock	AD	AS	r_t^e	Prices	Output Gap	Policy i_t
Productivity (temp $\hat{A}_t \uparrow$)	$\cdot$	$\downarrow$	$\downarrow$	$\downarrow$	-	$\downarrow$
Productivity (perm $\hat{A}_t, \hat{A}_{t+1} \uparrow$)	$\uparrow$	$\downarrow$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
Productivity news ($\hat{A}_{t+1} \uparrow$)	$\uparrow$	$\cdot$	$\uparrow$	$\uparrow$	+	$\uparrow$
Markup ($\hat{\mu}_t \uparrow$)	$\cdot$	$\uparrow$	$\cdot$	$\uparrow$	-	Trade-off
Public spending (temp $\hat{G}_t \uparrow$)	$\uparrow$	$\downarrow$	$\uparrow$	$\uparrow$	+	$\uparrow$
Preferences ($\hat{\xi}_t \downarrow$)	$\downarrow$	$\cdot$	$\downarrow$	$\downarrow$	-	$\downarrow$

Note: Dots indicate “no change”. Signs reflect movements from the *initial* equilibrium.

Parameter and Variable Map

Parameters

$\sigma > 0$	IES	$\eta > 0$	Inverse Frisch elasticity
$\beta \in (0, 1)$	Discount factor	$\kappa > 0$	Slope of the NKPC

Variables & Shocks

$\hat{Y}_t, \hat{Y}_t^n, \hat{Y}_t^e$	Log-deviation of Output (Actual, Natural, Efficient)
$\hat{i}_t, r_t^e$	Nominal Policy Rate vs. Wicksellian Real Rate
p^*	Long-run Price Anchor
$\hat{G}_t, \hat{A}_t$	Public Spending & Productivity Shocks
$\hat{\mu}_t, \hat{\xi}_t$	Markup & Preference Shocks